# BILIRUBUN METABOLISM

## Objectives

1. Outline the steps of heme catabolism (bilirubin formation).
2. List the products of heme catabolism and their fate
3. Mention the biochemical differences between unconjugated and conjugated bilirubin.
4. Define hyperbilirubinemia and jaundice

- **Bilirubin is the catabolic product of heme metabolism. Within the physiologic range, bilirubin has cytoprotective and beneficial metabolic effects, but at high levels it is potentially toxic**.

- **Hemes are cyclic tetrapyrroles that contain iron and are commonly found as the prosthetic group of hemoglobin, myoglobin and the cytochromes.**

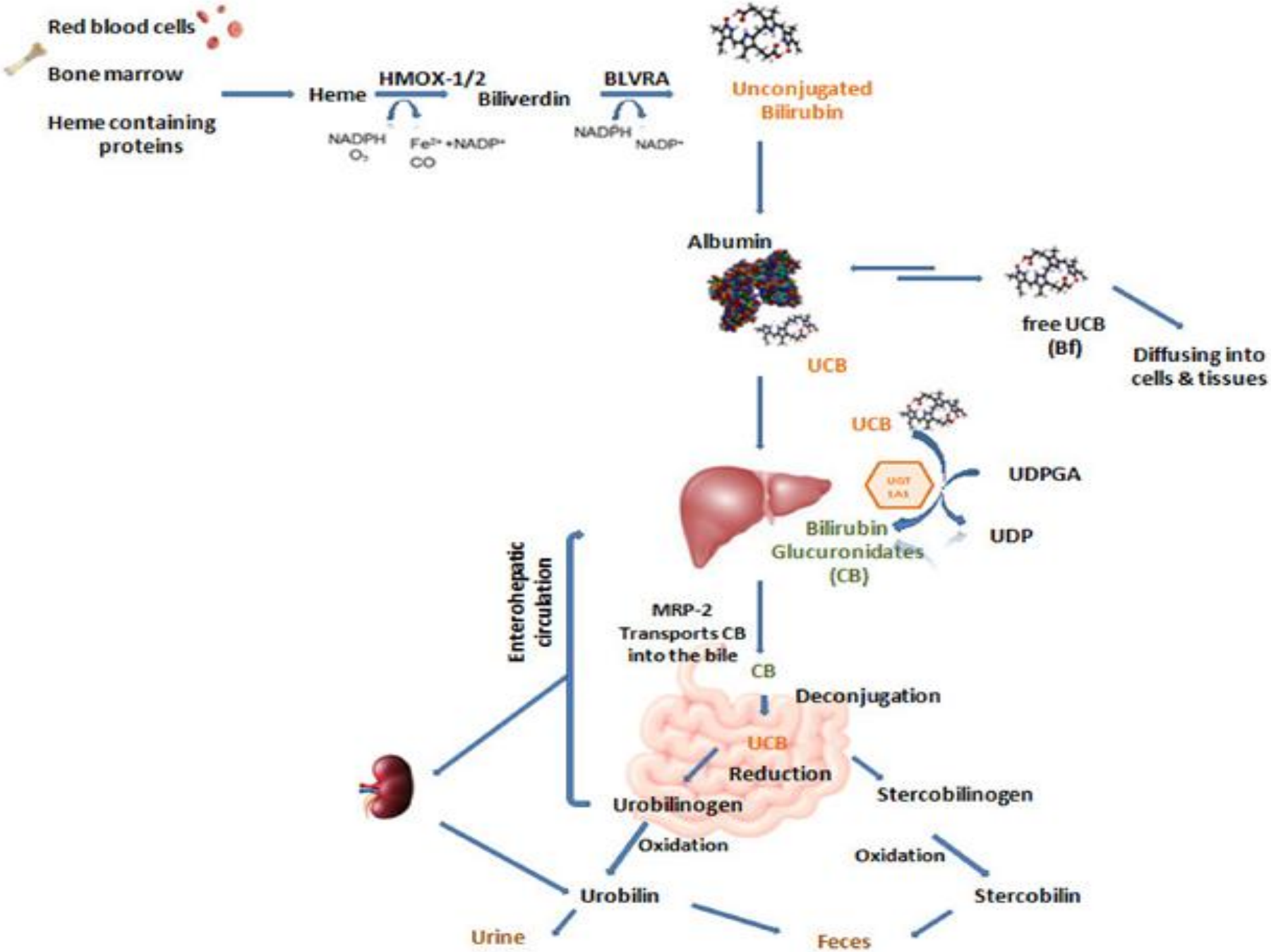
- **About 80% of degraded heme comes from old erythrocytes and 20% comes from premature erythrocytes in the bone marrow which are destroyed prior to release into the circulation and a minor component is derived from other cell types**

# Heme Degradation

- **Degradation begins inside macrophages of the spleen, which remove old and damaged (senescent) erythrocytes from circulation.**
- **RBCs are engulfed by cells of the reticuloendothelial system. The globin is recycled into amino acids, which are reused .**

- **Heme is oxidized; the heme ring is opened by heme oxygenase. The oxidation occurs on a specific carbon, producing the linear tetrapyrrole biliverdin, ferric iron ($Fe^{3+}$), and CO.**
- **In the next reaction, the second bridging methylene is reduced by biliverdin reductase, producing bilirubin. The green pigment is thus converted to the red-orange bilirubin.**

- **Bilirubin is then transported in the serum by albumin to the liver, where it is conjugated with glucuronate by bilirubin glucuronyl transferase and excreted in the bile.**
- **In the intestine, bilirubin is deconjugated and converted to urobilinogen and stercobilin.**
- **Some urobilinogen is reabsorbed and excreted as urobilin in the urine. Most urobilinogen is oxidized in the feces to stercobilin, which gives feces its color.**


Red blood cells
Bone marrow
Heme containing proteins
Heme
HMOX-1/2
NADPH
$O_2$
$Fe^{2+}$ +NADP+
CO
Biliverdin
BLVRA
NADPH
NADP+
Unconjugated Bilirubin
Albumin
UCB
free UCB (Bf)
Diffusing into cells & tissues
UCB
UDPGA
UDP
Bilirubin Glucuronidates (CB)
Enterohepatic circulation
MRP-2 Transports CB into the bile
CB
Deconjugation
UCB
Reduction
Urobilinogen
Stercobilinogen
Oxidation
Oxidation
Urobilin
Stercobilin
Urine
Feces

# Metabolism of Bilirubin

From hemoglobin breakdown to excretion

## 1. Hemoglobin Breakdown

→ Senescent RBCs destroyed in spleen
→ Hemoglobin
→ Heme + Globin

Hemoglobin → Heme + Globin

## 2. Formation of Unconjugated Bilirubin

Heme → Biliverdin → Unconjugated bilirubin

→ Heme
→ Biliverdin
→ Unconjugated bilirubin
→ Lipid-soluble
→ Transported bound to albumin

Albumin

## 3. Hepatic Uptake

→ Unconjugated bilirubin taken up by hepatocytes

## 4. Conjugation in Liver

→ Unconjugated bilirubin → glucuronic acid
→ Enzyme: UDP-glucuronyl transferase
→ Converts unconjugated → conjugated bilirubin
→ Water-soluble

## 5. Biliary Excretion

→ Conjugated bilirubin excreted into bile
→ Enters intestine

## 6. Intestinal Metabolism

→ Converted to urobilinogen by gut bacteria

## 7. Final Excretion

→ Urobilin
→ urine (yellow color)

→ Stercobilin
→ feces (brown color)

Disruption at any step leads to jaundice

## Heme Degradation Associated Diseases

- **In patients with abnormally high red cell lysis or obstructive liver damage, bilirubin can accumulate, leading to the clinical manifestation of jaundice**

# Jaundice

- **Jaundice, also known as**
- **hyperbilirubinemia, is defined as a yellow discoloration of the body tissue resulting from the accumulation of excess bilirubin.**
- **The normal serum levels of bilirubin are less than 1 milligram per deciliter (mg/dL).**

- **However, the clinical presentation of jaundice with peripheral yellowing of the eye sclera, also called scleral icterus, is best appreciated when serum bilirubin levels exceed 3 mg/dl.**
- **With further increase in serum bilirubin levels, the skin will progressively discolor ranging from lemon yellow to apple green, especially if the process is long-standing; the green color is due to biliverdin.**

# Types of Hypebilirubinemia

- **UNCONJUGATED HYPERBILIRUBINEMIA**
- **Bilirubin is not processed by the liver, water-insoluble, and binds to albumin to travel in the blood. It makes up >85% of total bilirubin in this condition.**
- **Causes:pre-hepatic: hemolytic anemia**
- **hepatic : Gilbert syndrome (reduced UGT enzyme activity) and Crigler-Najjar syndrome.**
- **Clinical Presentation**
- **Jaundice, normal urine color. Often seen in neonates due to physiological breakdown of red blood cells.**

# CONJUGATED HYPERBILIRUBINEMIA

**Bilirubin is processed by the liver (glucuronidation), making it water-soluble, allowing it to be excreted in bile.**

- **Causes: hepatic: (viral hepatitis, alcoholic liver disease).**
- **post-hepatic:gallstones (cholelithiasis), tumors, or pancreaticcancer.**
- **Clinical Presentation:**
- **Jaundice, dark ("coca-cola") urine, and pale or clay-colored stools.**

- **Potential Toxicity:**
- **Unconjugated: High Unconjugated (Indirect) bilirubin levels can cross the blood-brain barrier and cause kernicterus in infants.**
- **Conjugated:High conjugated (direct) bilirubin Usually does not cause neurotoxicity**

# Gilbert syndrome and Crigler-Najjar syndrome types I and II

- hereditary, nonhemolytic, unconjugated hyperbilirubinemias that result from mutations in the UDP-glucuronosyltransferase gene that impair bilirubin conjugation.
- Gilbert syndrome:
- UDP-glucuronosyltransferase enzyme activity is 10-30% of normal, resulting in mild unconjugated hyperbilirubinemia with bilirubin levels below 5 mg/dL

- **Crigler-Najjar types I and II syndromes**
- **inherited autosomal recessive disorders that result from mutations in UDP-glucuronosyltransferase gene**
- **Crigler-Najjar type I:**
- **UDP-glucuronosyltransferase enzyme activity is absent, unconjugated bilirubin levels of 20-45 mg/dL. infants with Crigler-Najjar type I syndrome usually develop bilirubin encephalopathy (Kernicterus) with seizures and may die from severe neurologic diseases in the absence of treatment**

- **Crigler-Najjar type II:**
- **Individuals with have less than 10% of norma UDP-glucuronosyltransferase activity, resulting in bilirubin levels of 5-25 mg/dL. Kernicterus is rare in Crigler-Najjar type II, but patients with increased bilirubin levels (greater than 15 mg/dL) have a high chance of its occurrence**

- **1. What is the enzyme that catalyzes the first, rate-limiting step in heme catabolism?**
- A) Bilirubin reductase
  B) Heme synthetase
  C) **Heme oxygenase**
  D) Uroporphyrinogen synthase
- **2. The initial product formed from the breakdown of heme by heme oxygenase is:**A) Bilirubin
  B) Urobilin
  C) **Biliverdin**
  D) Ferritin

- **3. Where does the initial breakdown of heme from senescent red blood cells primarily occur?**
  A) Bone marrow
  B) Liver hepatocytes
  C) **Reticuloendothelial System (Spleen, Liver, Bone Marrow)**
  D) Kidney tubules

- **4. What happens to the iron ($Fe^{2+}$) released from heme during catabolism?**
  A) Excreted in urine
  B) Converted to biliverdin
  C) **Bound to transferrin and transported to bone marrow for reuse**
  D) Stored as ferritin in the liver

Good
Luck